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Tutorial on Implicit Neural Representations for Medical Imaging



HARVARD
UNIVERSITY

MICCAI 2024
Marrakech
MOROCCO



UNIVERSITY
OF TWENTE.



mcm

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Jonathan R.
Schwartz

Benedict Wiestler

Björn Menze

Tutorial on
Implicit Neural Representations for Medical Imaging

Tutorial Schedule

Introduction (08.00 - 08.40) (Julian):

- *What's an INR and what are Neural Fields?*
- *What do we need to train INRs - and why would we want to train them in the first place?*
- *Formal Definitions & Intuitions for Medical Imaging*

Practical Coding (8.40 - 09.20) (Nil):

- *It's easy! How to train your first INR - and what's different in comparison to classical deep learning applications*
- *Using some hacks to make INRs work*
- *Demoing a real-world INR application in k-space*

In-depth Guest Lecture (9.20 - 10.00) (Jelmer):

- *Applications of INRs in registration*
- *Applications of INRs in shape modeling*

Coffee Break (10.00 - 10.30)

Overview of INR applications and domains (10.30-11.00) (Supro, Veronika) :

- *Applications in Undersampling, Reconstruction and Rendering*
- *Applications in Segmentation, Registration*
- *Applications in Temporal Dynamics and Progression Modeling*

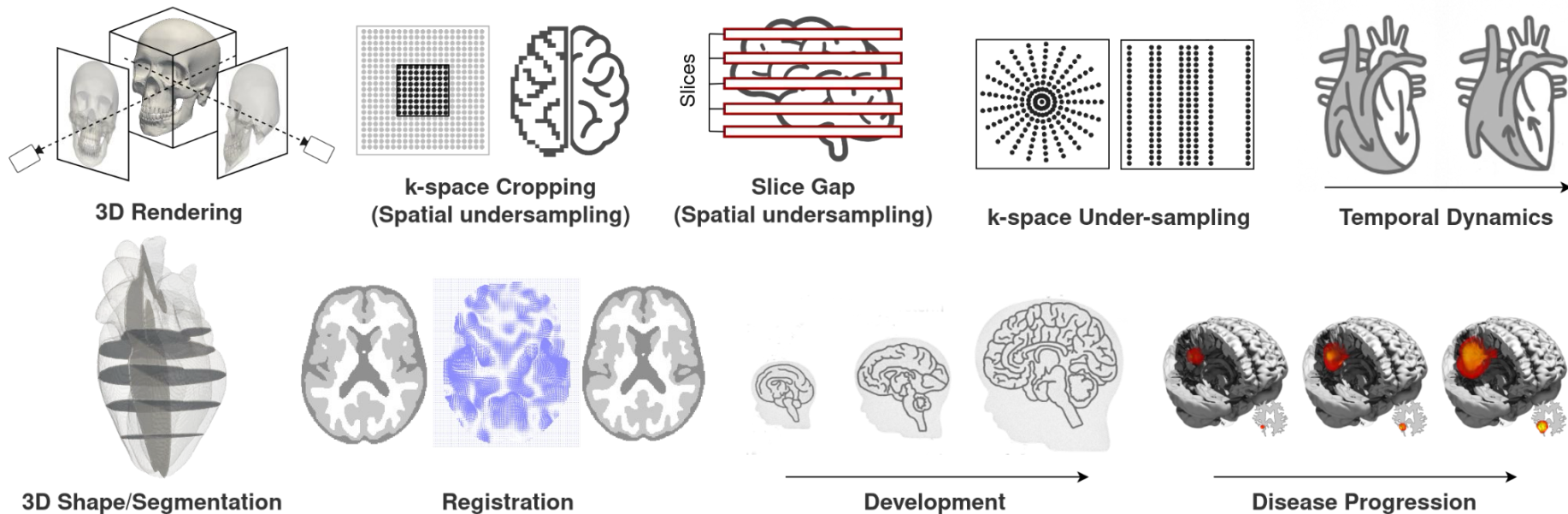
Guest Lecture on Advanced Concepts of INRs (11.00-11.50) (Jonathan):

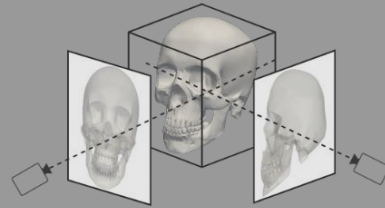
- *Cohort-based Learning and Meta Learning for Initialization*
- *INRs for Compression and From Data to Funct*

Cohort Learning, Neural Rendering and Conclusion (11.50 - 12h30) (Maik, Vasiliki):

- *An application of cohort-based learning in fetal imaging*
- *Neural Rendering and Frameworks to start your research*
- *Future Research Directions & Conclusion*

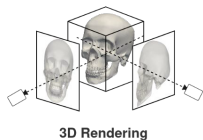
INRs: Continuity & Prior across Tasks





3D Rendering

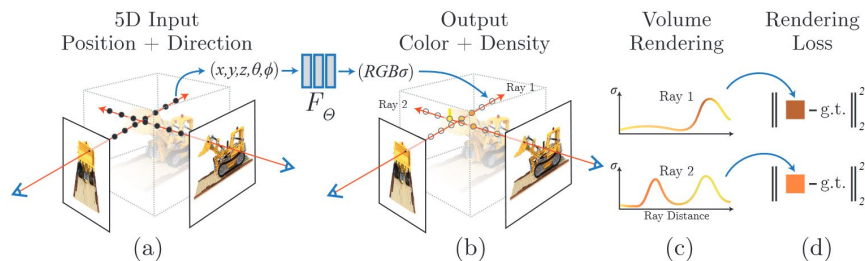
Image rendering



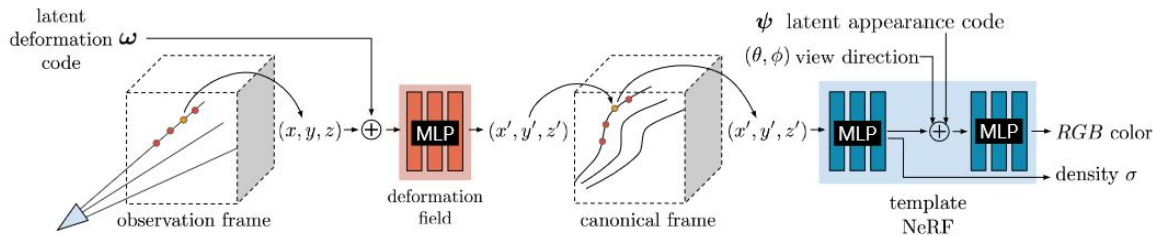
NeRFs: Short Reminder

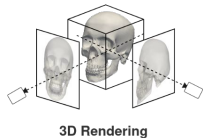
3D volume from 2D slices

NeRF



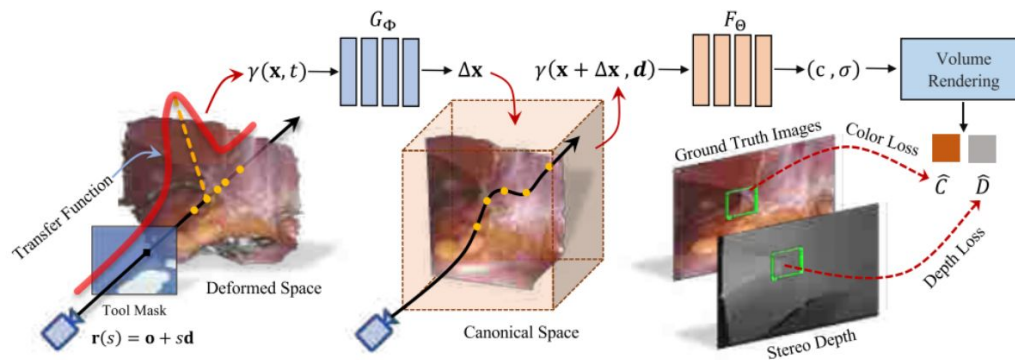
Nerfies





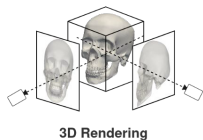
Neural Rendering in a Surgical Setting

3D volume from 2D slices



Additional Challenges/Adaptations:

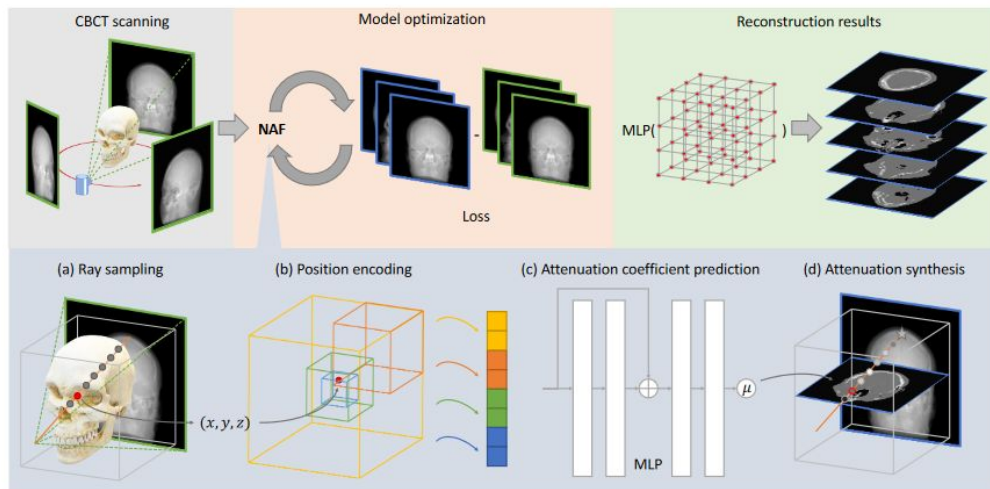
- Occlusion through surgical tools $\rightarrow$ tool masking $\rightarrow$ avoid shooting through tool pixels
- Increase sampling around tissue surface (transfer function with stereo depth) for better 3D structures



CBCT

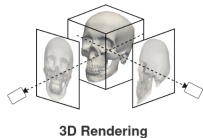
Neural Rendering in CBCT

3D from 2D slices



Prediction of attenuation coefficients

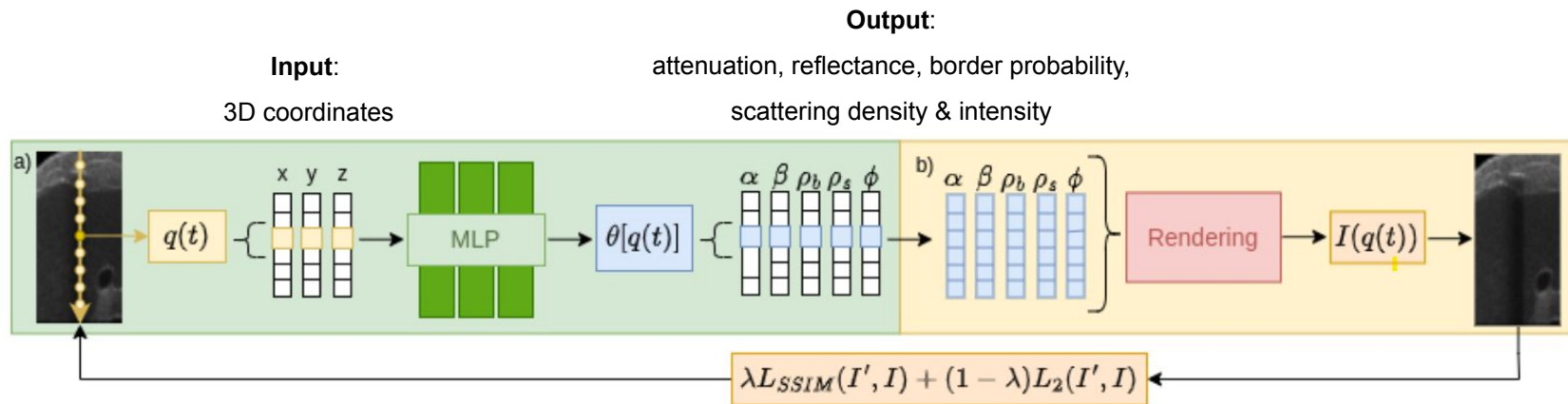
Zha, Ruyi; Zhang, Yanhao; Li, Hongdong (2022): NAF: Neural Attenuation Fields for Sparse-View CBCT Reconstruction: Springer Nature Switzerland (Lecture Notes in Computer Science).



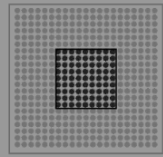
US

Neural Rendering from Ultrasound Images

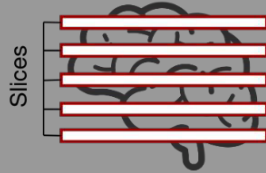
Inclusion of a physics-based US volume rendering model - representing reflected & backscattered energy



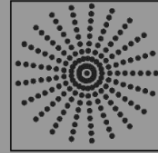
$$I(r, t) = I_0 \cdot \prod_{n=0}^{t-1} [(1 - \beta(r, n)) \cdot G(r, n)] \cdot \exp(-\int_{n=0}^{t-1} (\alpha \cdot f \cdot dt))$$



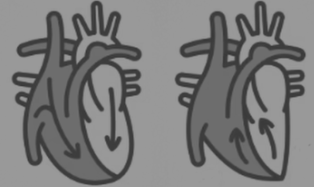
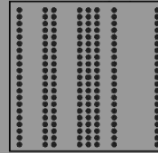
k-space Cropping
(Spatial undersampling)



Slice Gap
(Spatial undersampling)



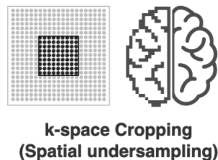
k-space Under-sampling



Temporal Dynamics

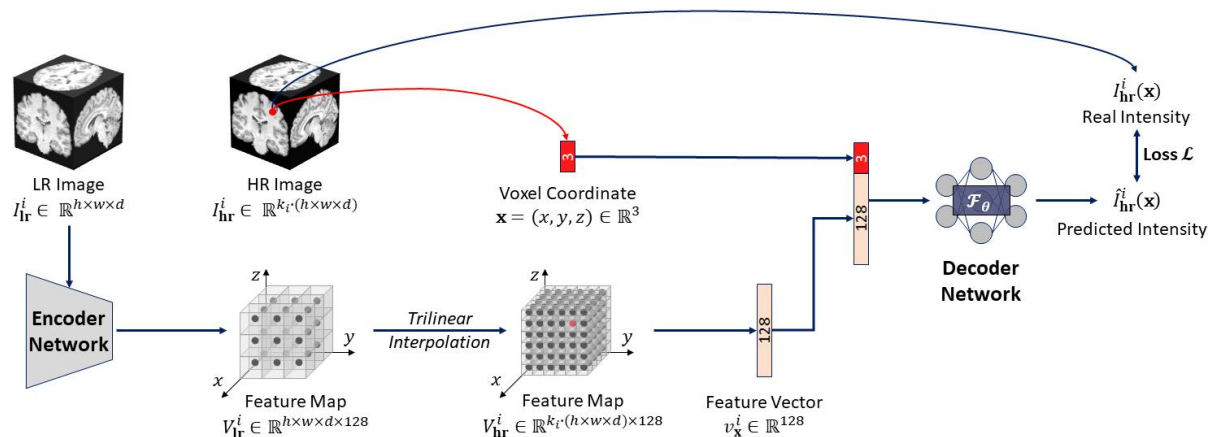
Mainly in image domain

Image reconstruction - Part 1

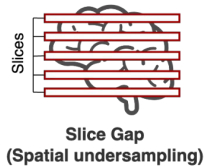


Super-Resolution in Image Reconstruction

Increase spatial resolution within slices/volume



- Simple interpolation network $\rightarrow$ Image-to-Image
- Independent of modality since it works on already reconstructed LR
 $\rightarrow$ **BUT no data consistency with sensor domain ensured**

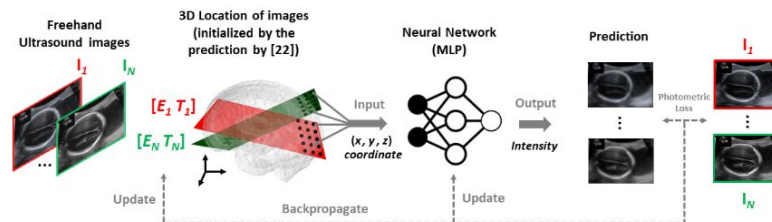


US

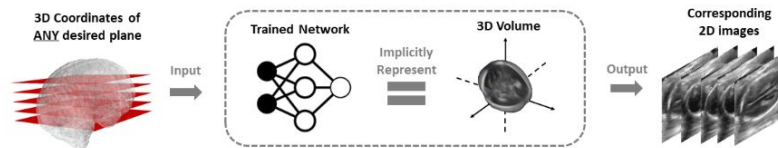
Slice-to-Volume Reconstruction

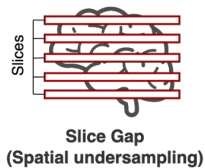
Create an isotropic volume from cross-sectional slices

Train a Neural Network to Implicitly Represent a 3D Volume



Use the Trained Network as a Continuous 3D Volume during Inference

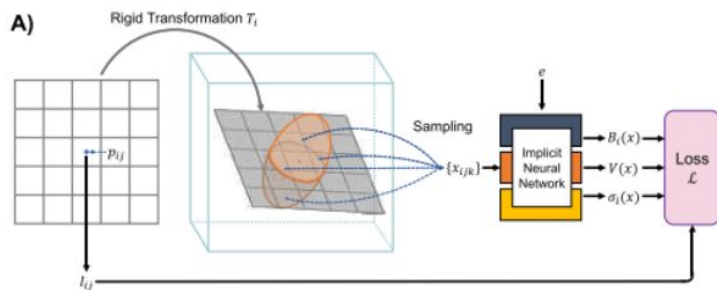




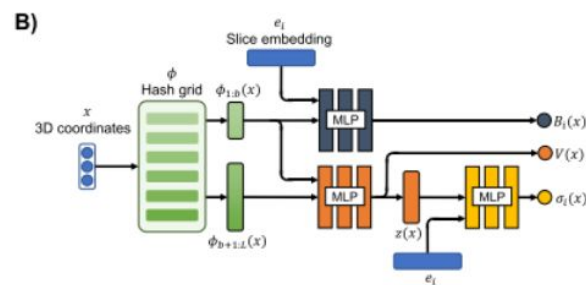
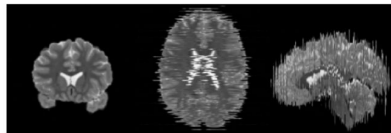
MRI

Slice-to-Volume Reconstruction

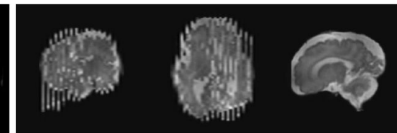
Additionally account for motion in between slices



Simulated adult brain data

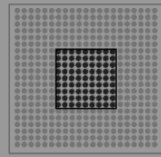


Simulated fetal brain data

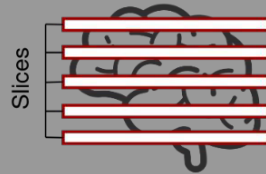


Additional Challenges/Adaptations:

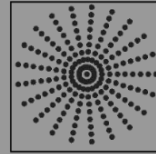
- Motion between individual slices -> include transformations, bias fields and PSF
- Outlier removal using pixel-level and slice-level variances



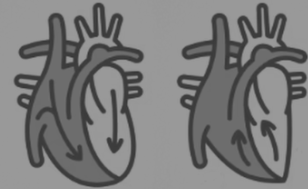
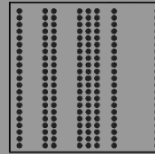
k-space Cropping
(Spatial undersampling)



Slices
Slice Gap
(Spatial undersampling)



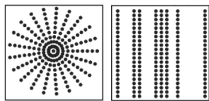
k-space Under-sampling



Temporal Dynamics

Mainly including sensor domain

Image reconstruction - Part 2



k-space Under-sampling

CT/MR

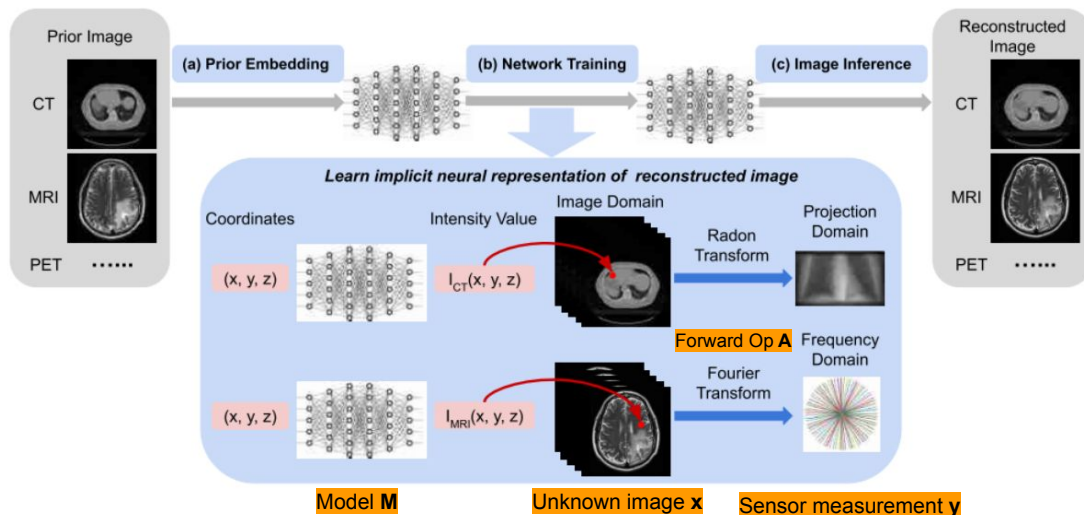
General Concept: Sensor Domain-based Image Recon:

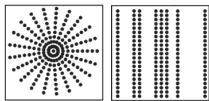
Inverse Problem

$$y = Ax + e$$

Optimize INRs for the Inverse Problem

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \|A\mathcal{M}_{\theta} - y\|_2^2, \quad x^* = \mathcal{M}_{\theta^*}$$



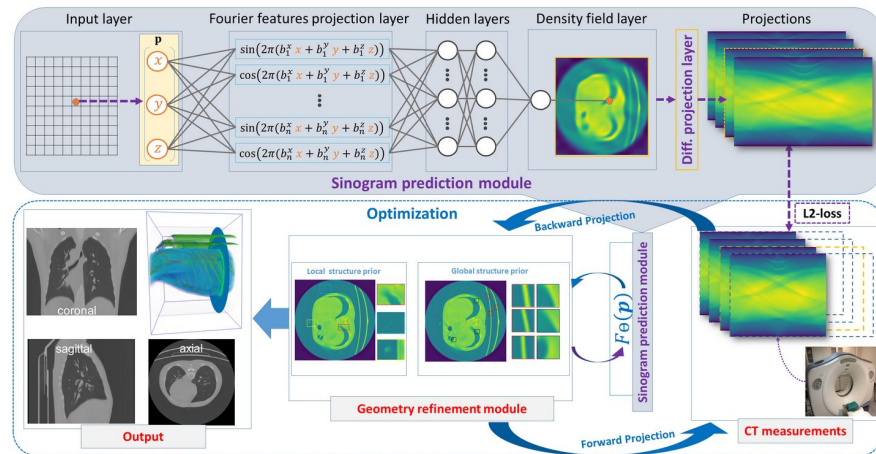


k-space Under-sampling

CT

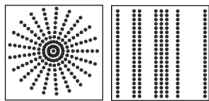
Sensor-based Image Recon: CT Example

Predict underlying density fields from incomplete/degraded CT sinograms



Additional Adaptations:

- Geometric refinement with local/non-local geometric priors (Regularization)

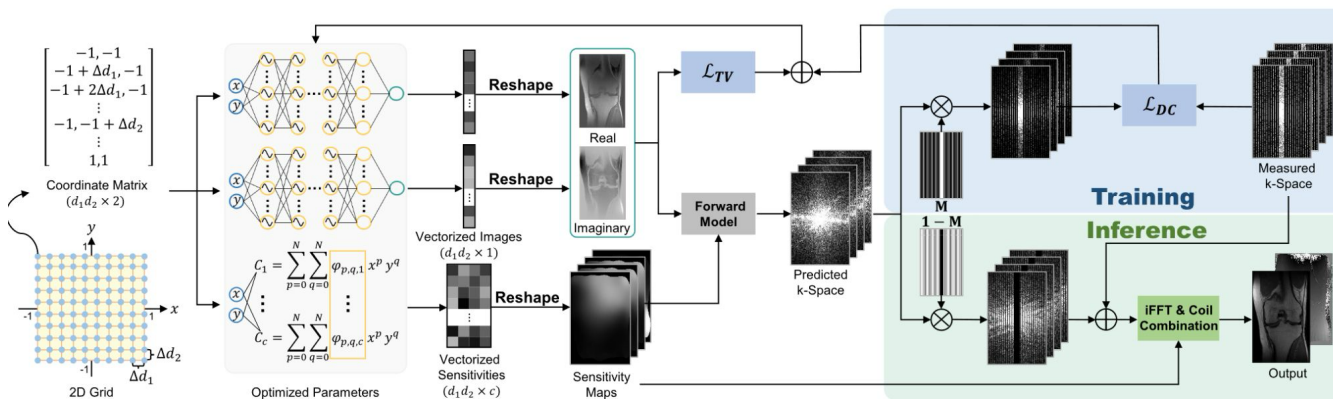


k-space Under-sampling

MR

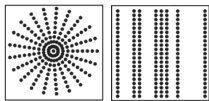
Sensor-based Image Recon: MR Example

Predict k-space lines that have not been acquired



Additional Challenges:

- MR images are complex-valued and acquired with different coils $\rightarrow$ multiple/separate INR predictions which feed into forward model

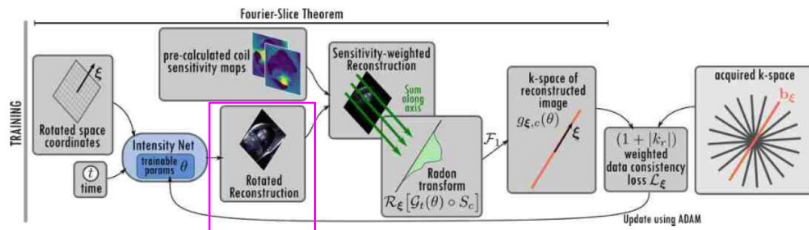


k-space Under-sampling

MR

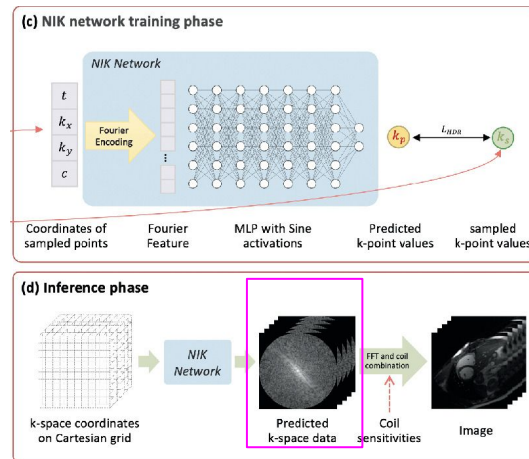
Two Different Training Strategies in MR

Images are predicted and then forward model is applied to images to compare to sensor domain images to compare to sensor domain



Input: Image coordinates
Output: **Image intensities**
Loss: k-space

Interpolation is conducted in sensor domain and image is obtained at inference



Input: k-space coordinates
Output: **k-space intensities**
Loss: k-space

Spatial Undersampling: Temporal Dynamics

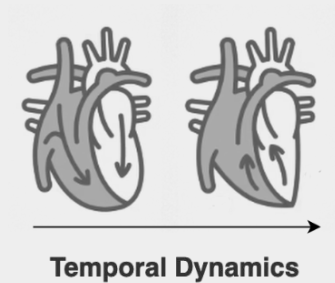
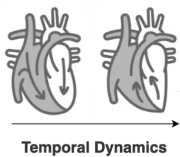
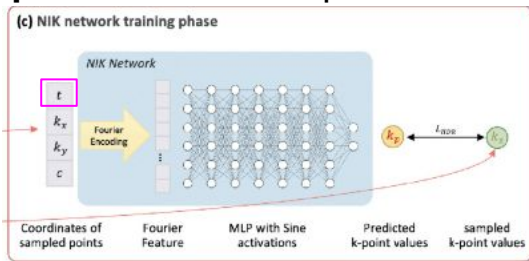


Image reconstruction



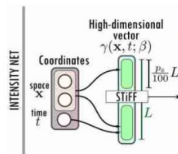
Including the extra temporal dimension

a) **Implicit:** Additional input to INR

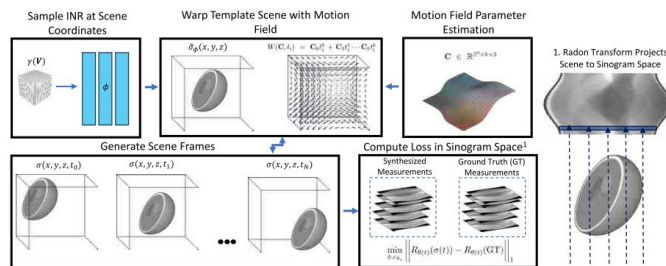


Different Encoding Strategies

- Fourier Features
- Hash Encoding
- Spatiotemporal Feature Encoding



b) **Explicit:** Motion field estimates applied to reference

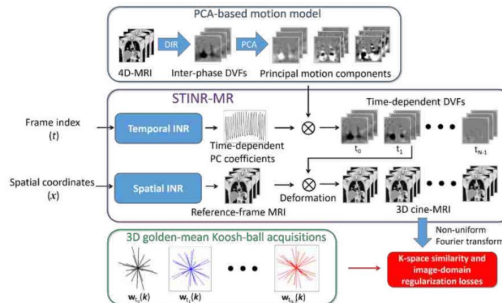


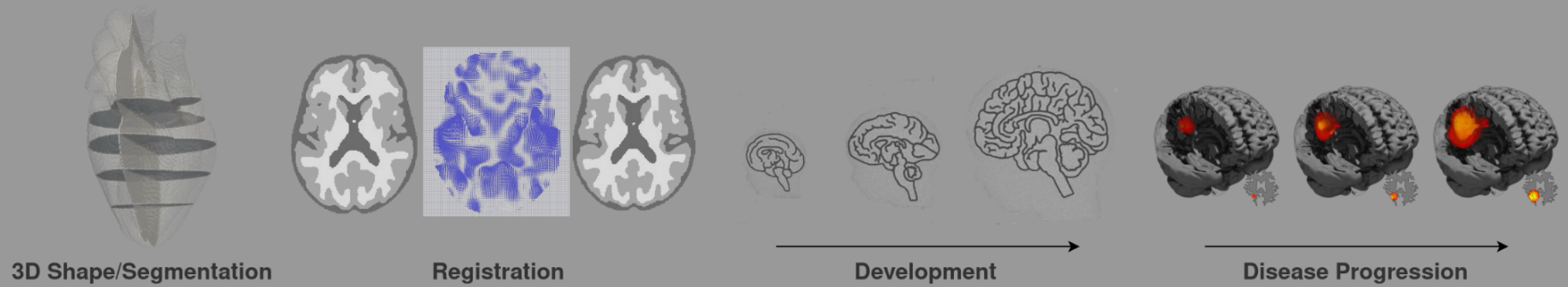
4D-CT

Parametrization of motion fields + summation over time

3D cine MRI

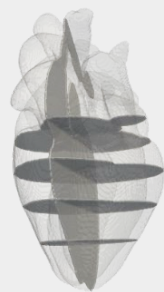
Create motion bases + combine them based on INR-predicted coefficient





Beyond Rendering & Reconstruction ...

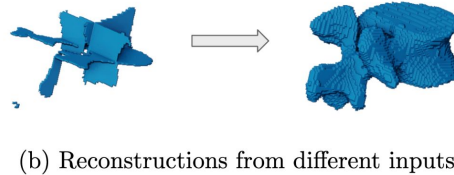
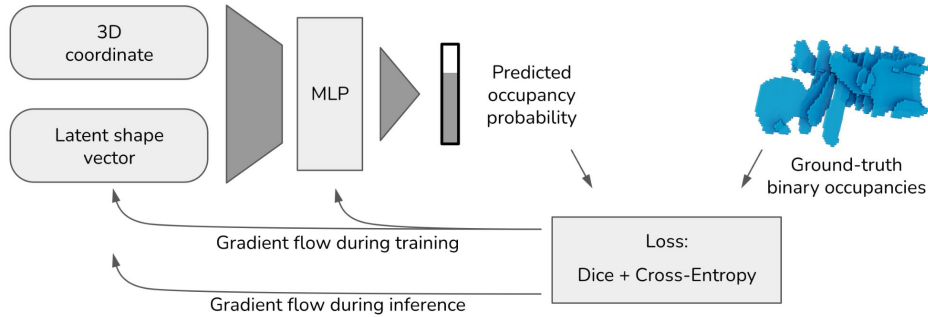
Shape From Segmentation



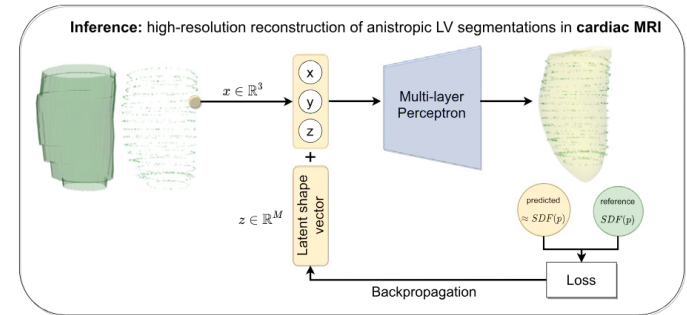
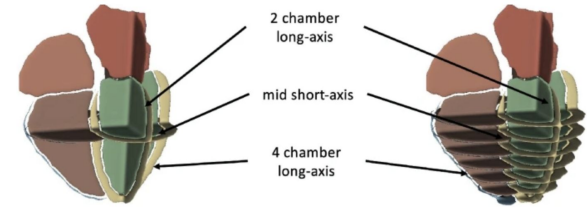
3D Shape/Segmentation

3D Shape Prediction

Shape Completion



Applications:



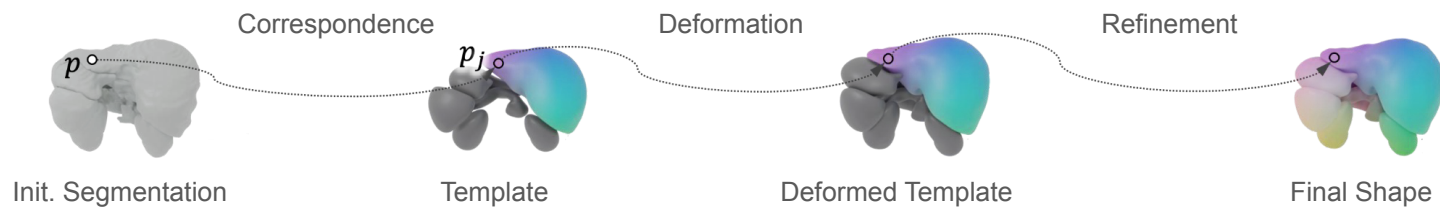
Amiranashvili, Tamaz, et al. "Learning shape reconstruction from sparse measurements with neural implicit functions." *International Conference on Medical Imaging with Deep Learning*. PMLR, 2022.

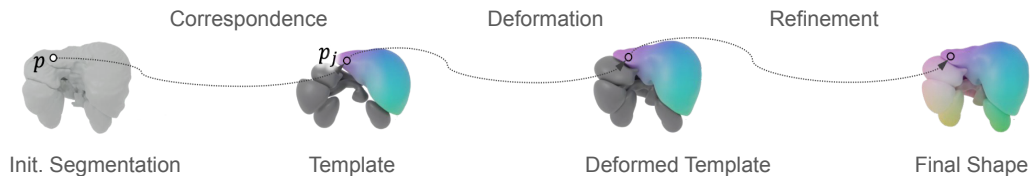
Amiranashvili, Tamaz, et al. "Learning continuous shape priors from sparse data with neural implicit functions." *Medical Image Analysis* 94 (2024): 103099.

Muffoletto, Marica, et al. "Neural Implicit Functions for 3D Shape Reconstruction from Standard Cardiovascular Magnetic Resonance Views." *International Workshop on Statistical Atlases and Computational Models of the Heart*. 2023.

Sander, Jörg, et al. "Reconstruction and completion of high-resolution 3D cardiac shapes using anisotropic CMRI segmentations and continuous implicit neural representations." *Computers in Biology and Medicine* 164 (2023): 107266.

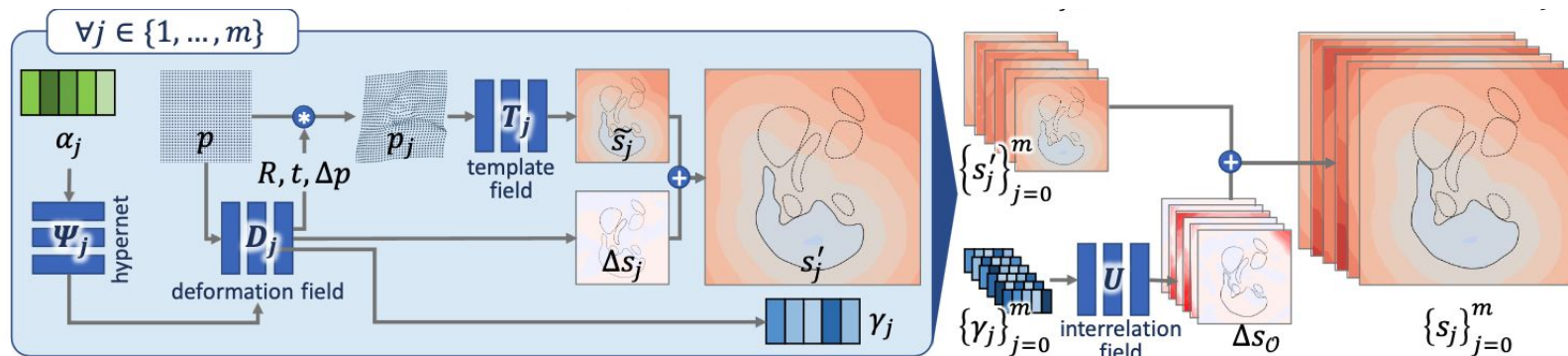
Shape Reconstruction





Hypernetwork

Shape Reconstruction



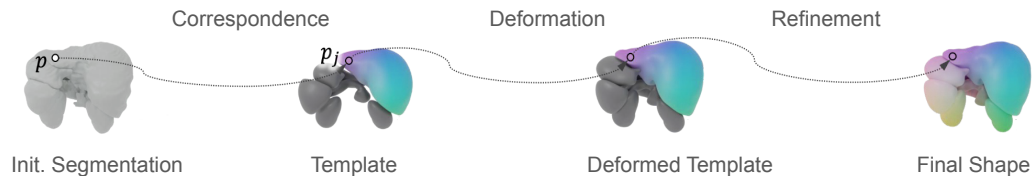
Key Adaptations:

- Hypernetwork to predict instance-specific weights, that produces deformation field
- Shared template field for each object category

Zhang, Minghui, et al. "Implicit Shape Modeling for Anatomical Structure Refinement of Volumetric Medical Images." arXiv preprint arXiv:2312.06164 (2023).

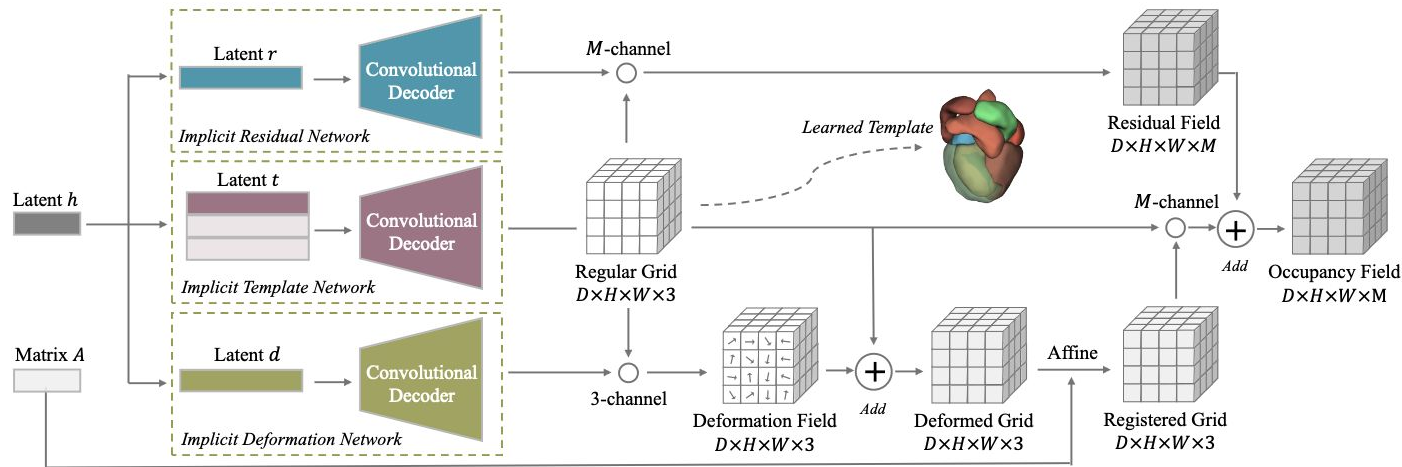
Deng, Yu, Jialong Yang, and Xin Tong. "Deformed implicit field: Modeling 3d shapes with learned dense correspondence." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2021.

Liu, Yuchun, et al. "Implicit Modeling of Non-rigid Objects with Cross-Category Signals." Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 38. No. 4. 2024.



Auto-Decoder
+ Feature
Interpolation

Shape Reconstruction



Key Adaptations:

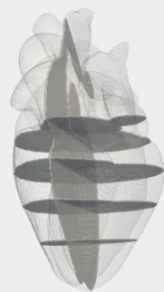
- Use of spatially correlated latent and feature interpolation for learned template

Yang, Jiancheng, et al. "Generating Anatomically Accurate Heart Structures via Neural Implicit Fields." International Conference on Medical Image Computing and Computer-Assisted Intervention. 2024.

Yang, Jiancheng, et al. "ImplicitAtlas: learning deformable shape templates in medical imaging." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022.

Sun, S., Han, K., Kong, D., Tang, H., Yan, X., & Xie, X. (2022). Topology-preserving shape reconstruction and registration via neural diffeomorphic flow. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 20845-20855).

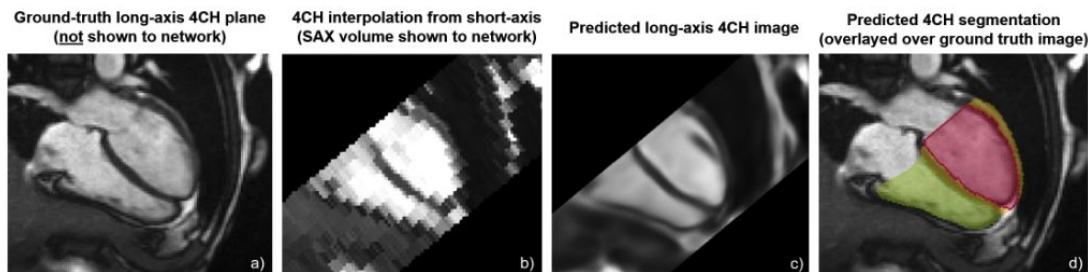
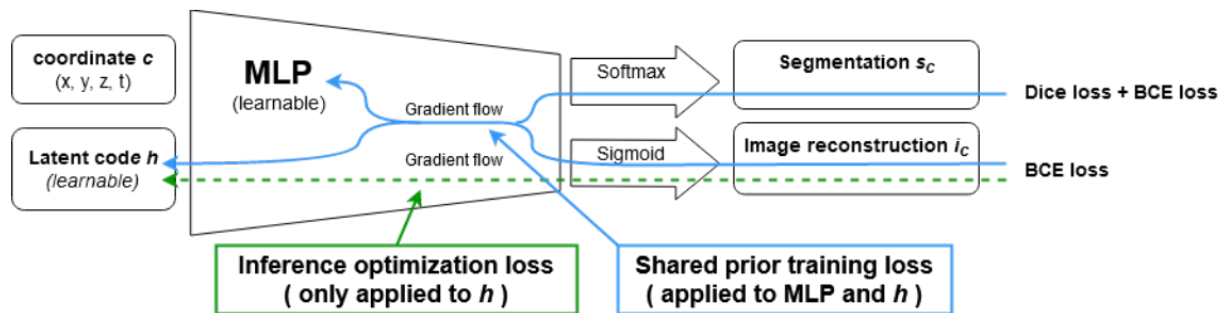
Shape From Image (Segmentation)



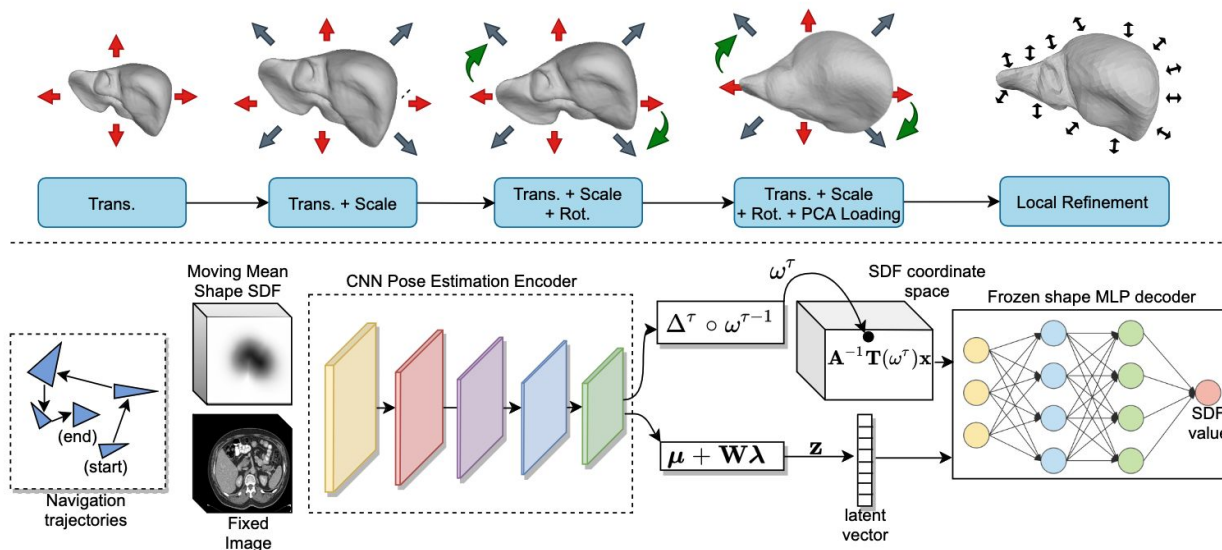
3D Shape/Segmentation

3D Shape Prediction

Segmentation



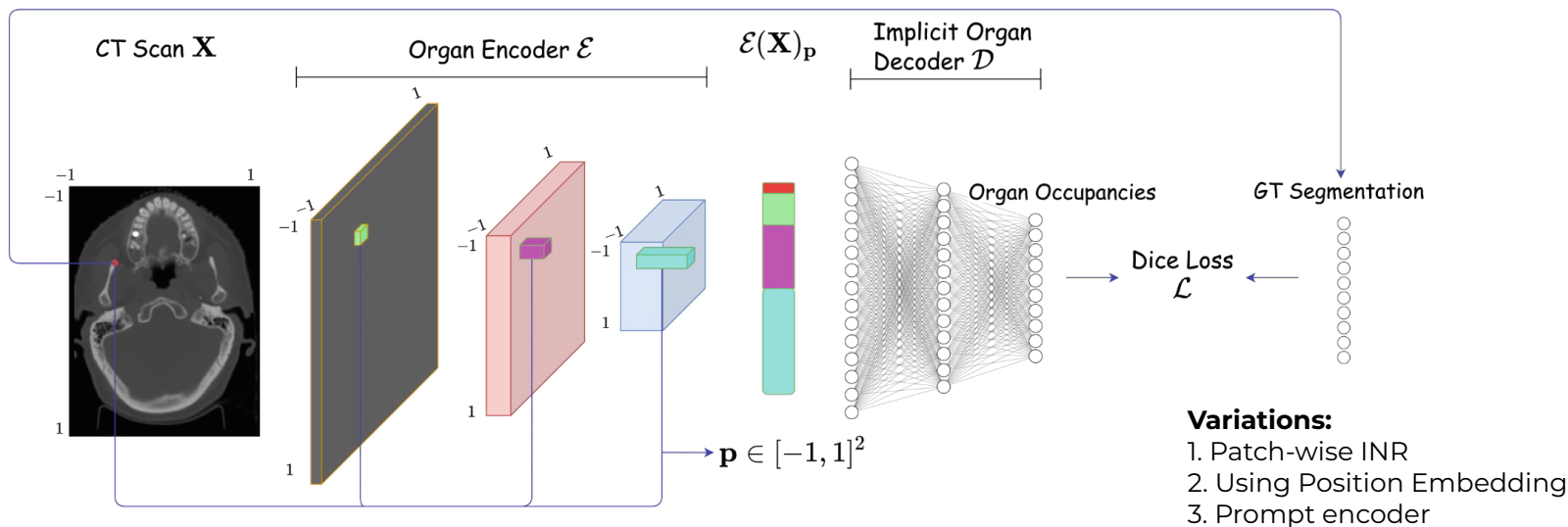
Segmentation



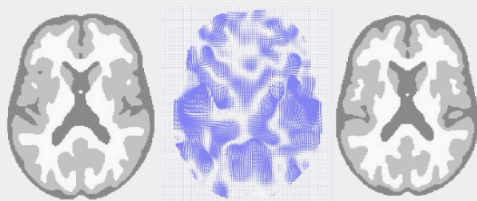
Key Adaptations:

- Rigid and non-rigid pose estimation followed by refinement

Segmentation

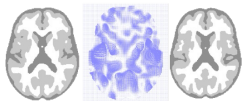


Multimodal and Diffeomorphic

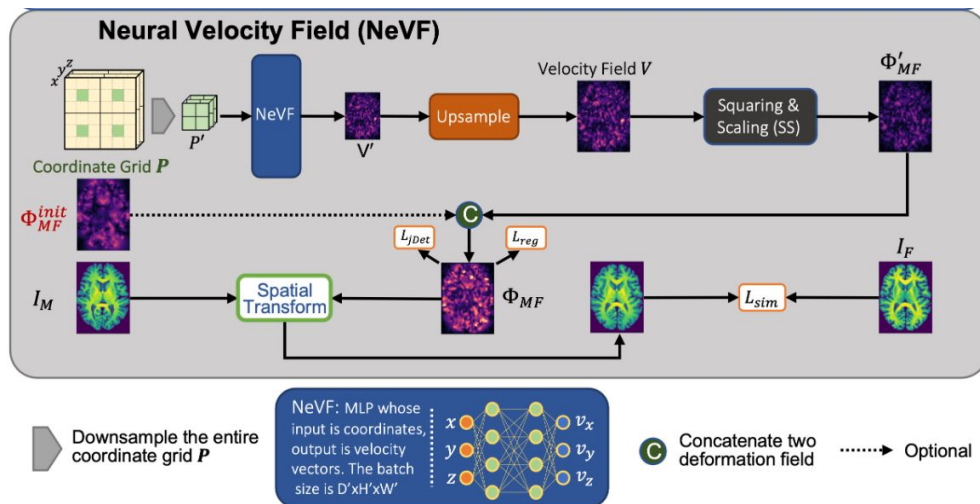


Registration

Registration

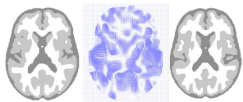


Registration

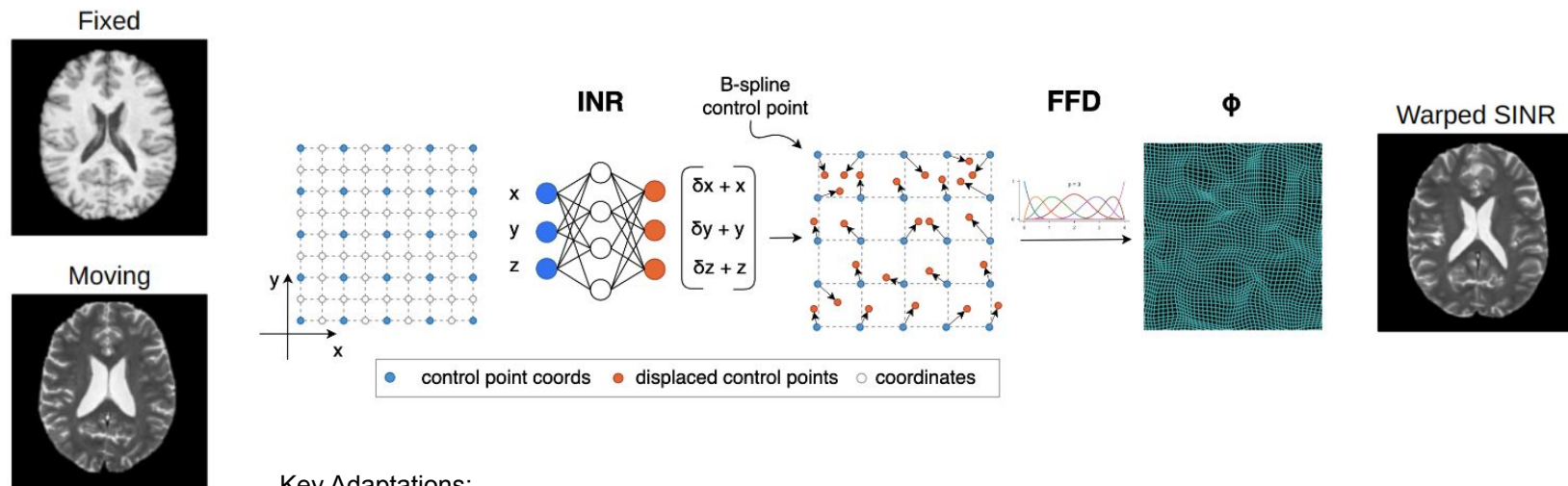


Key Adaptations:

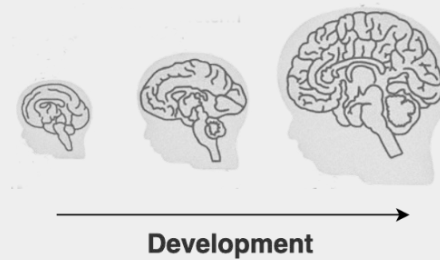
- Continuous velocity field $\rightarrow$ more accurate displacement via integration
- Loss, that requires spatial gradients is easy to compute via auto-differentiation



Registration



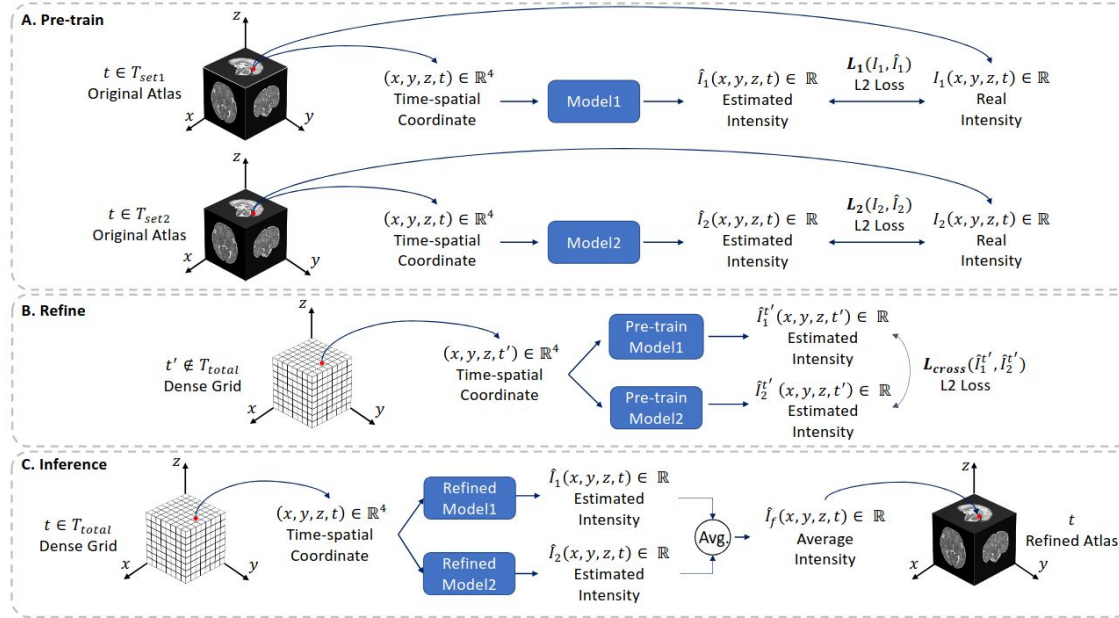
Fetal/Neonatal Brain MRI



Growth Modelling

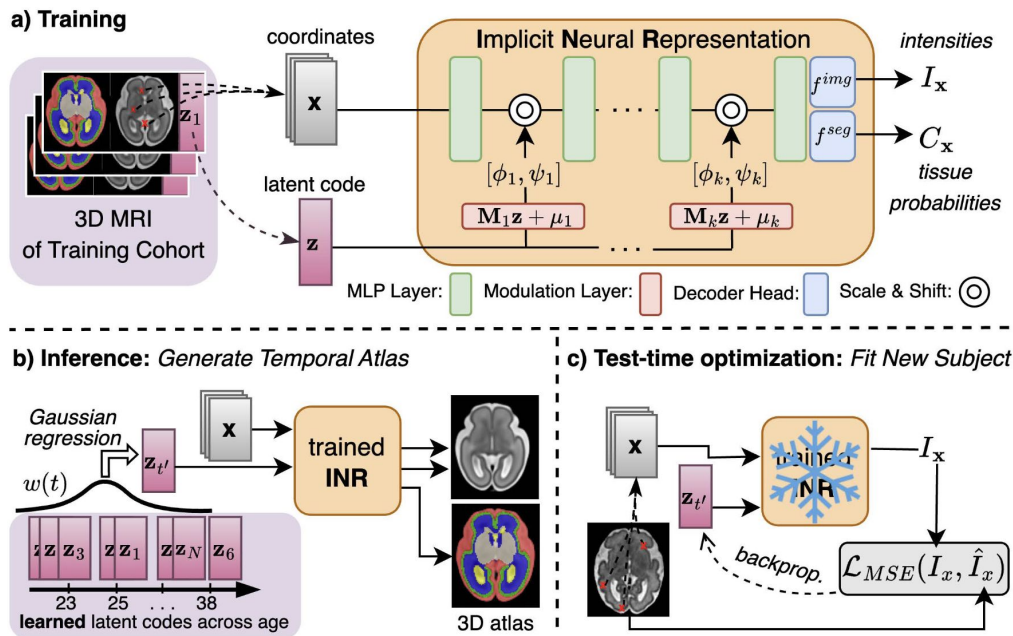


Explicit Temporal Modeling

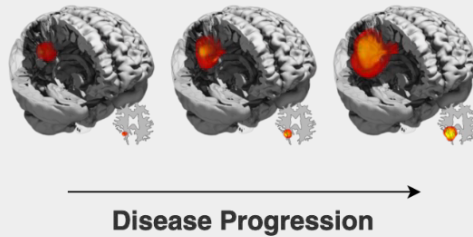




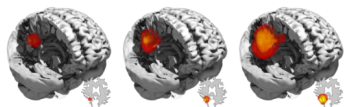
Implicit Temporal Modeling



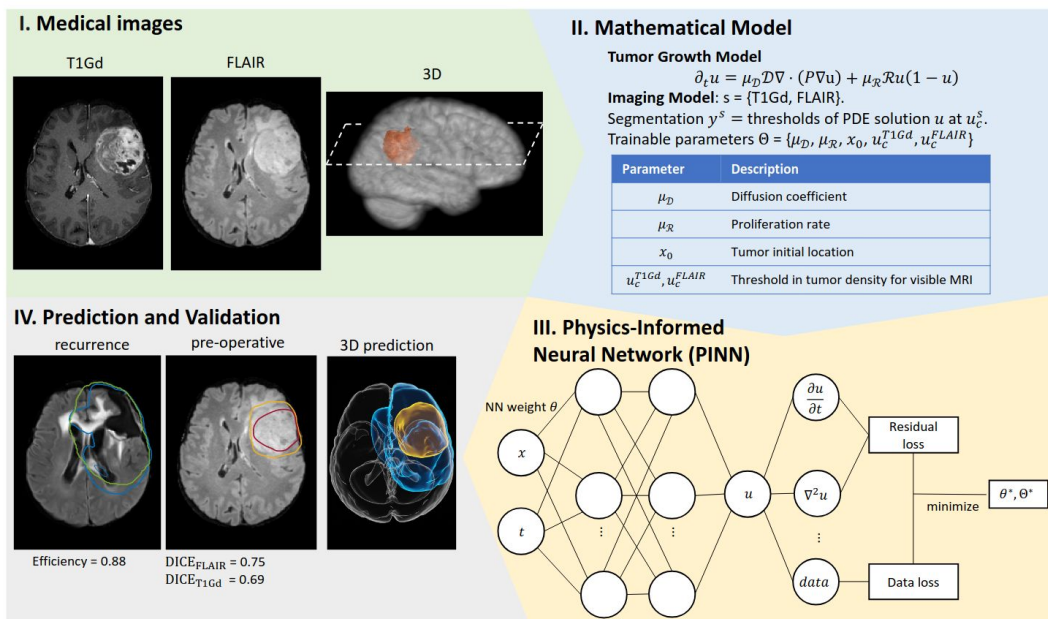
Partial Differential Equations (PDEs)



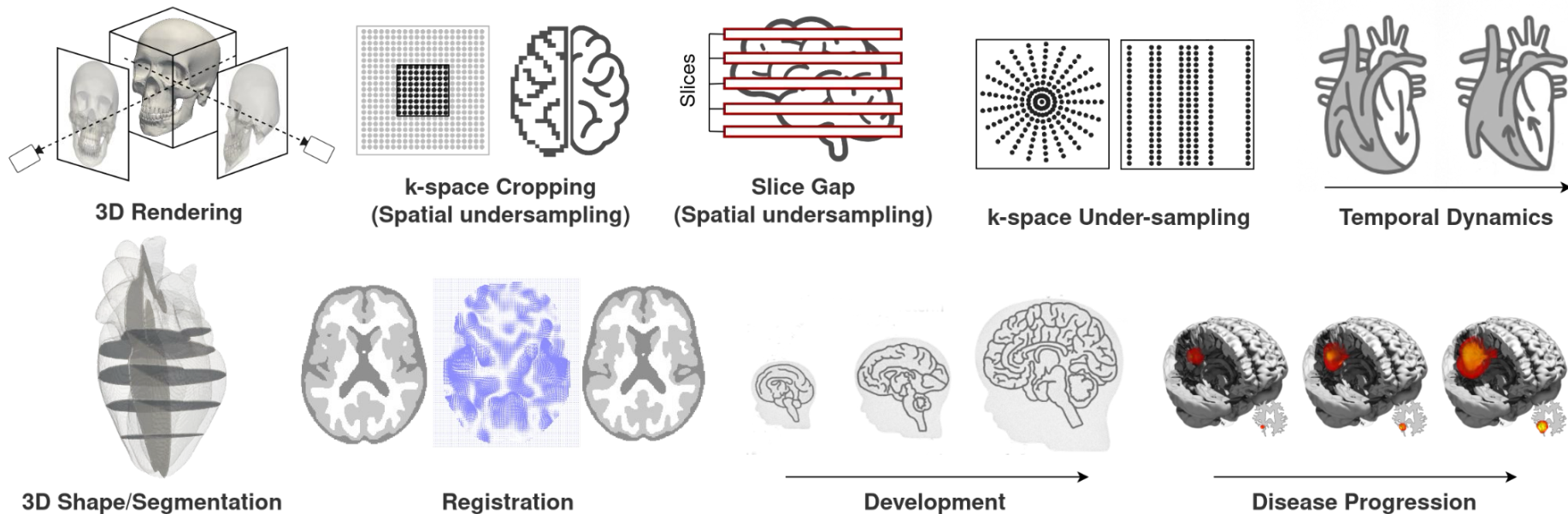
Disease Progression



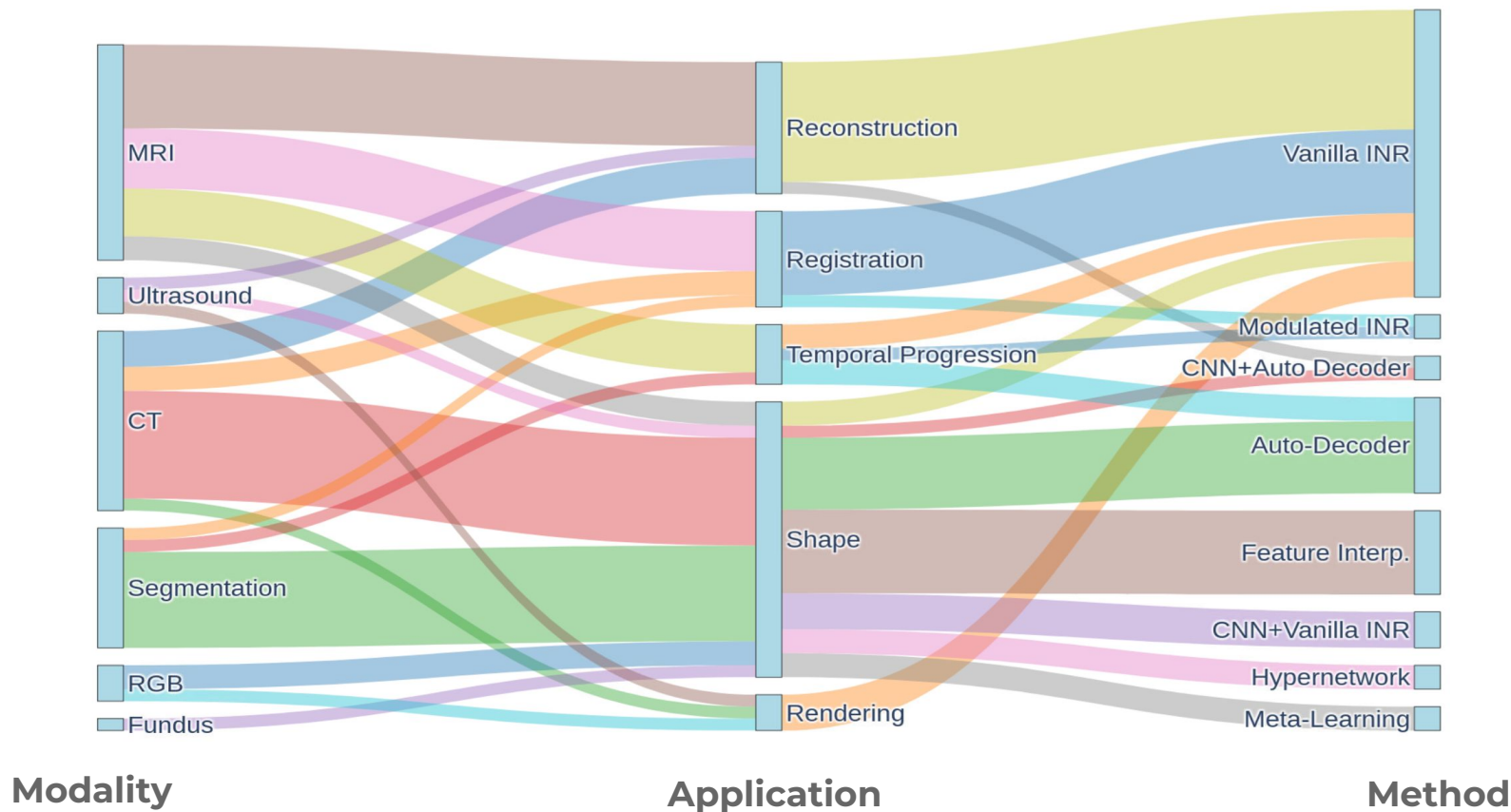
Brain Tumor Growth Modelling



INRs: Continuity & Prior across Tasks



INRs: Medical Imaging Landscape so far ...



Data Stored as Function

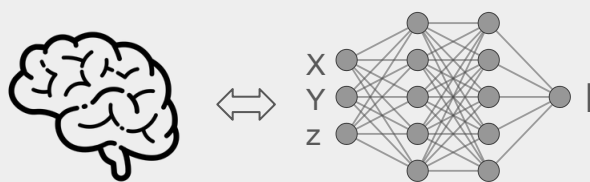
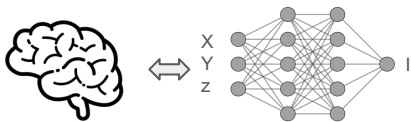
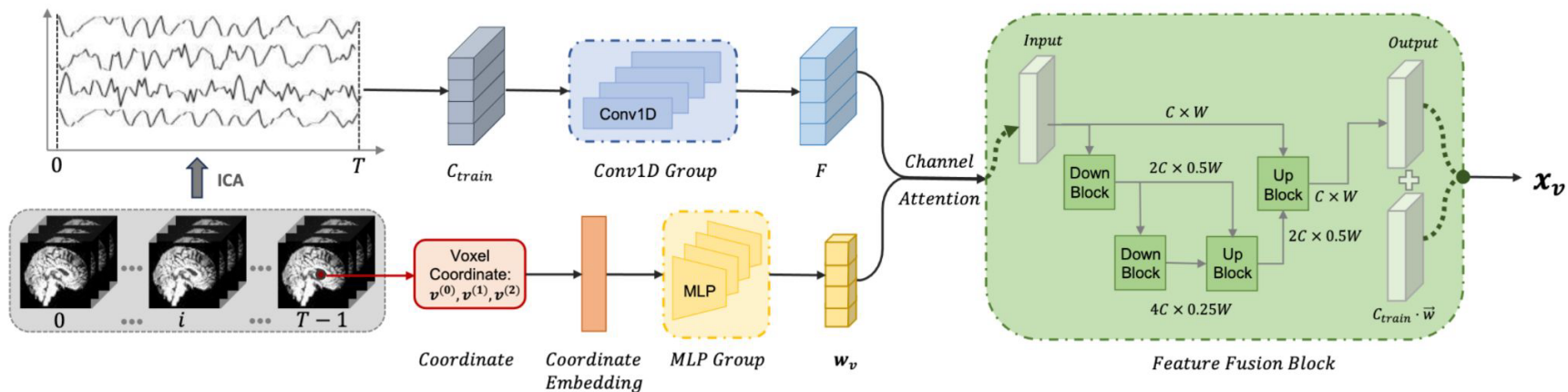


Image Compression



fMRI

Image Compression



Key Adaptations:

- Linear compression model and reuse of neurons to reduce inter-region redundancies

Advanced Strategies for INR Training

- **Include Uncertainty:** Multiple MLPs to predict distribution of recon [Vasconcelos_2023]
- **Enhance High-frequency details:** Iterative recon with hidden feature propagation [Sawdayee_2023]
- **Regularization:** include k-space domain specific properties for INR regularization [Spieker 2024]
- **INR+GAN:**
 - Multi-sequence MRI translation using shift modulated INR [Chen 2024, MELBA]
 - GRAF: generator predicts patches, discriminator compares to patch from real image [Corona-Figueroa 2022]
- **INR+Diffusion:** Super-resolution of 3D microscopy Image with diffusion prior [Lee 2024, MICCAI]
- **Disentangle 3D shape from 2d views: Extend reconstruction dimensions:**
 - **MR Multi-contrast:** Train with two LR contrasts and have to separate output heads [McGinnis 2023]
- and many more....

Checkout this page: <https://github.com/xmindflow/Awesome-Implicit-Neural-Representations-in-Medical-imaging>

Vasconcelos, Francisca; He, Bobby; Singh, Nalini; Teh, Yee Whye (2023): UncertalNR: Uncertainty Quantification of End-to-End Implicit Neural Representations for Computed Tomography.
 Sawdayee, Haim; Vaxman, Amir; Bermano, Amit H. (2023): OReX: Object Reconstruction from Planar Cross-sections Using Neural Fields. In: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR): IEEE, S. 20854–20862.
 Spieker, Veronika; Eichhorn, Hannah; Stelter, Jonathan K.; et al.: Self-Supervised k-Space Regularization for Motion-Resolved Abdominal MRI Using Neural Implicit k-Space Representation – MICCAI 2024. MICCAI 2024.
 Corona-Figueroa, Abril; Frawley, Jonathan; Bond-Taylor, Sam; Bethapudi, Sarath; Shum, Hubert P. H.; Willcocks, Chris G. (2022): MedNeRF: Medical Neural Radiance Fields for Reconstructing 3D-aware CT-Projections from a Single X-ray.
 McGinnis, J. et al. (2023). Single-subject Multi-contrast MRI Super-resolution via Implicit Neural Representations. In: Greenspan, H., et al. Medical Image Computing and Computer Assisted Intervention – MICCAI 2023. MICCAI 2023.

Questions/Comments?